



C#

Level 1

ProgressBar



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Definition

ProgressBar Control

 A Windows Forms control used to display the progress of an operation, usually from 0% to 100%.

 In this program, it works with a Timer and buttons to Start, Pause, Resume, Stop, and Reset the progress.

Explanation

Pause State

 `isPaused` is a Boolean variable used to track whether the progress is paused.

Timer

 `timer1` runs every 500 milliseconds and increases the ProgressBar value by 1.

ProgressBar Configuration

 Minimum = 0, Maximum = 100, and Value starts at 0.

 The label displays the current ProgressBar value as a percentage.

Procedure

1. Set `isPaused` to `false`.
2. Set the Timer interval to 500 milliseconds.
3. Configure the initial button states.
4. Set the ProgressBar range from 0 to 100.
5. Start the Timer when Start is clicked.
6. Stop and restart the Timer with Pause and Resume.
7. Stop the Timer with Stop without changing the current value.
8. Reset the ProgressBar and label to 0%.

💡 Example

```
if (progressBar1.Value < progressBar1.Maximum)
{
    progressBar1.Value++;
    label1.Text = progressBar1.Value + "%";
}
```

➡ Result

0%

1%

2%

3%

...

99%

100%

Note

- ◆ `Minimum` defines the lowest value.
- ◆ `Maximum` defines the highest value.
- ◆ `Value` represents the current progress.
- ◆ `Interval = 500` means the Timer ticks approximately every 500 milliseconds.



Comparison



Start

- ✓ Starts the Timer and progress.



Pause / Resume

- ✓ Pause stops the Timer.
- ✓ Resume starts it again from the current value.



Stop / Reset

- ✓ Stop pauses the progress without resetting the value.
- ✓ Reset stops the Timer and returns the progress to 0%.

⊘ Warning

- ◆ The ProgressBar cannot exceed its `Maximum`.
- ◆ The condition `Value < Maximum` prevents the value from going above 100.
- ◆ When the value reaches 100, the Timer stops.

✦ Summary

- ✓ ProgressBar → displays the current progress.
- ✓ Timer → controls how often the progress increases.
- ✓ `Value` → stores the current progress.
- ✓ `Start()` → starts the Timer.
- ✓ `Stop()` → stops the Timer.
- ✓ `isPaused` → tracks the pause state.

Thank you 😊